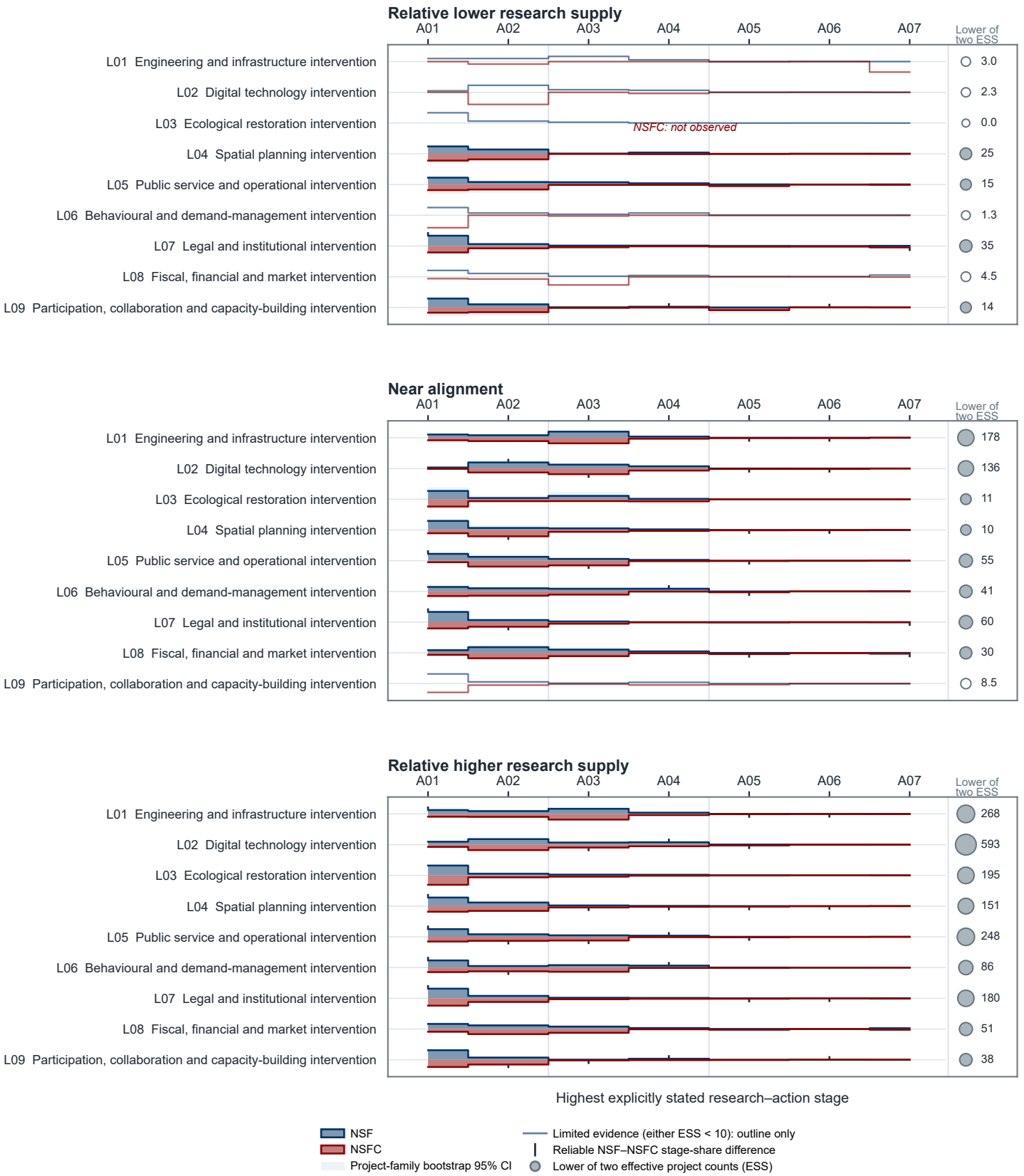
b Stated action-stage boundaries across intervention modes

Highest explicitly stated action stage after formal-Target and award-time entropy balancing

Count-weighted CLR mismatch classes (thresholds −0.5, +0.5); each observed seven-stage profile sums to 100%



A01 Foundational knowledge stage · A02 Method development stage · A03 Controlled validation stage · A04 Real-world validation stage

A05 Decision and operation integration stage · A06 Diffusion and scaling stage · A07 Post-implementation evaluation stage

Reliable differences: both ESS ≥ 10, 95% interval excludes zero, and BH-FDR q < 0.05 across 189 cells.

Stage refers to the highest action explicitly stated in project text; it does not establish real-world implementation or impact.